

Module 3 Report

Molly Szeman

The Roux Institute, Northeastern University

ALY6040: Data Mining

Prof. Umesh Hodeghatta

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I. Code Walk Through

The .txt file of the “Seeds” dataset was read into R. The .txt file did not contain column names, so the variable names were entered from the information provided on the UC Irvine Machine Learning Repository. According to the dataset description, the data is made up of seeds of three different types of wheat, Kama, Rosa and Canadian, with 70 entries for each type, for the purpose of clustering analysis.

The data contains 210 rows and 7 columns. (The original data contained an additional column that identified the cluster, and this column was dropped). All the columns are continuous quantitative ratio variables that are measurements of characteristics of seeds. The variables are Area, Perimeter, Compactness, Length of Kernel, Width of Kernel, Asymmetry Coefficient, Length of Kernel Groove. The data is complete with no missing values (Figure 1).

Figure 1: Data Summary of Seeds Dataset

```
> skim(data)
```

Data Summary										
Name	data									
Number of rows	210									
Number of columns	7									
Column type frequency:										
numeric	7									
Group variables	None									
Variable type: numeric										
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
1 Area	0	1	14.8	2.91	10.6	12.3	14.4	17.3	21.2	
2 Perimeter	0	1	14.6	1.31	12.4	13.4	14.3	15.7	17.2	
3 Compactness	0	1	0.871	0.0236	0.808	0.857	0.873	0.888	0.918	
4 length_of_kernel	0	1	5.63	0.443	4.90	5.26	5.52	5.98	6.68	
5 width_of_kernel	0	1	3.26	0.378	2.63	2.94	3.24	3.56	4.03	
6 asymmetry_coefficient	0	1	3.70	1.50	0.765	2.56	3.60	4.77	8.46	
7 length_of_kernel_groove	0	1	5.41	0.491	4.52	5.04	5.22	5.88	6.55	

After initial analysis of the data, the next step was to standardize the data using the `scale()` function. Standardizing the data is critical because it prevents variables with larger scales (like Area, which ranges from 10.6 to 21.2) from having a greater influence over the cluster than variables with smaller scales (such as Compactness, which ranges from 0.808 to 0.918). As shown in Figure 2, after standardization using the `scale()` function, the mean of all variables is near 0, and the standard deviation of all variables is 1. This was achieved by subtracting the mean and dividing by the standard deviation. With the data in the standardized form, it was used to create clusters via the k-means and hierarchical methods.

Figure 2: Data Summary of Seeds Dataset after Standardization

```
> skim(data2)
```

— Data Summary —

Name	data2
Number of rows	210
Number of columns	7

Column type frequency:

numeric	7
---------	---

Group variables

None

— Variable type: numeric —

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
1 Area	0	1	-2.26e-15	1	-1.46	-0.886	-0.169	0.845	2.18	
2 Perimeter	0	1	-2.65e-15	1	-1.65	-0.849	-0.183	0.885	2.06	
3 Compactness	0	1	1.27e-15	1	-2.66	-0.597	0.104	0.710	2.00	
4 length_of_kernel	0	1	-4.13e-15	1.00	-1.65	-0.827	-0.237	0.793	2.36	
5 width_of_kernel	0	1	-2.45e-16	1	-1.66	-0.833	-0.0572	0.803	2.05	
6 asymmetry_coefficient	0	1	3.62e-16	1	-1.95	-0.757	-0.0673	0.711	3.16	
7 length_of_kernel_groove	0	1	-1.56e-16	1.00	-1.81	-0.739	-0.377	0.954	2.32	

```
> |
```

II. Analysis

Using the kmeans function, a k-means model was created to group the data into 4 clusters by generating 25 initial configurations. The k-means method creates clusters by creating groups with minimal intracluster mean square error. As a starting point, Model 1 was developed using four clusters and 25 configurations. The clusters were plotted using the fviz_nbclust function.

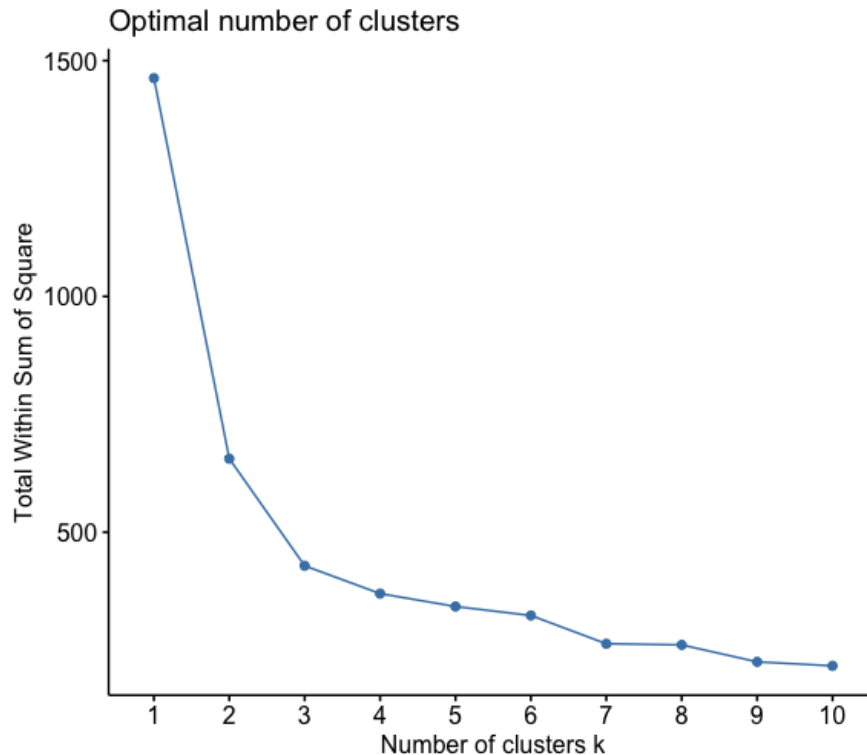
Figure 3: Cluster Plot of Model 1



Model 1, the initial k-means model, created four clusters of size 51, 65, 30, and 64. The within cluster sum of squares for each group were 86.3, 122.6, 49.8, and 110.7 (Appendix A). Dividing the distance between the sum of squares by the total sum of squares results in an explained variance of 74.7%, meaning 74.7% of the variance in the data is explained by the clustering. The results of Model 1 can be used to create additional models.

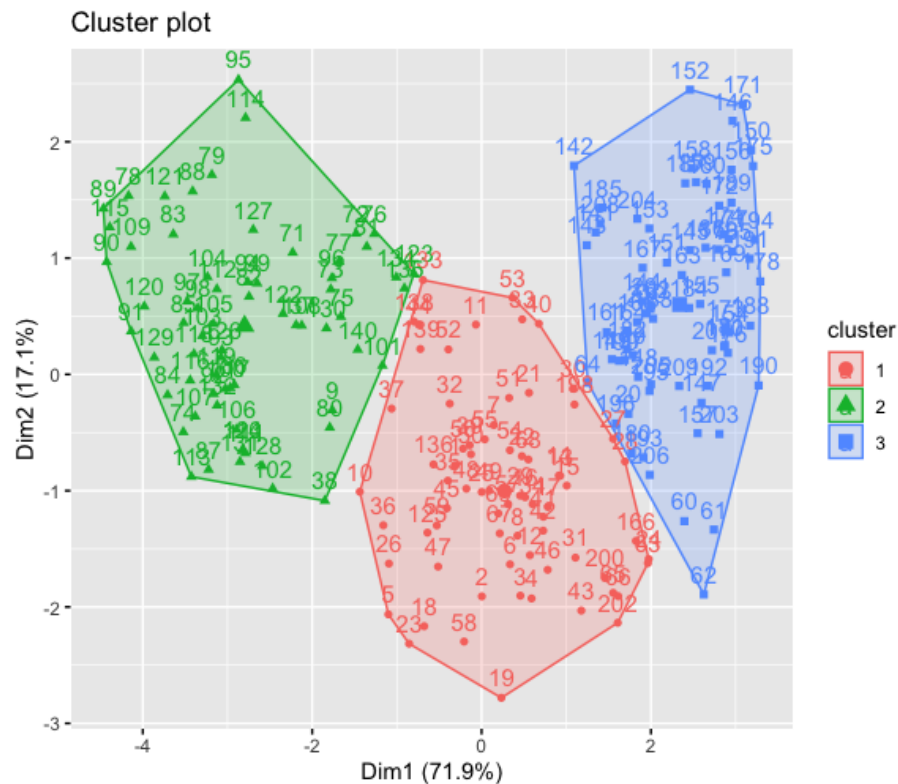
Using the Elbow method, the optimal number of clusters was determined by plotting the number of clusters (k) versus the total within the sum of squares. Figure 4 shows a steep decline in the total within the sum of squares between where $k=1$ and $k=2$, a less steep but still significant decline between $k=2$ and $k=3$, and after $k=3$, the total within the sum of squares levels off. According to the Elbow method, $k=3$ is the optimal number of clusters to use in the model, since the plot levels off after this point.

Figure 4: Optimal Number of Clusters for Model 1, Elbow Method



The k-means method was repeated for $k=3$ to build another model (Model 2) to cluster the data. Model 2 contains 3 clusters with sizes of 71, 67, and 72. Model 2 created more evenly divided groups than Model 1. The within cluster sum of squares for each group were 144.5, 139.6, and 144.6 (Appendix B). Dividing the distance between the sum of squares by the total sum of squares results in 70.7%, which is a lower explained variance than Model 1. Model 1 has 4 clusters compared to Model 2's 3 clusters, and adding clusters increases the explained variance.

Figure 5: Cluster Plot of Model 2



Additionally, a model was built using the hierarchical method and $k=3$, resulting in a dendrogram displaying the 3 distinct clusters (Figure 6). The euclidean method was used for distance between elements of the same group because it is calculating the distance between numerical values, and is calculated by finding the square root of the sum of the squared differences between the two vectors. For Model 3, average linkage was used to the distance between clusters, and for Model 3.1, complete linkage was used. The clusters were plotted using the `fviz_dend` function.

Figures 6 and 7 show the two dendrograms create similar models of clusters, although in Model 3, the groups are more evenly split and in Model 3.1, group 3 (green) is significantly larger. In Model 3, the cut height is 3, and in Model 3.1, the cut height is near 6, meaning there is higher dissimilarity between the clusters. This follows that the method used (complete linkage) uses the maximum distance between clusters.

Figure 6: Cluster Dendrogram of Model 3 (Average Linkage)

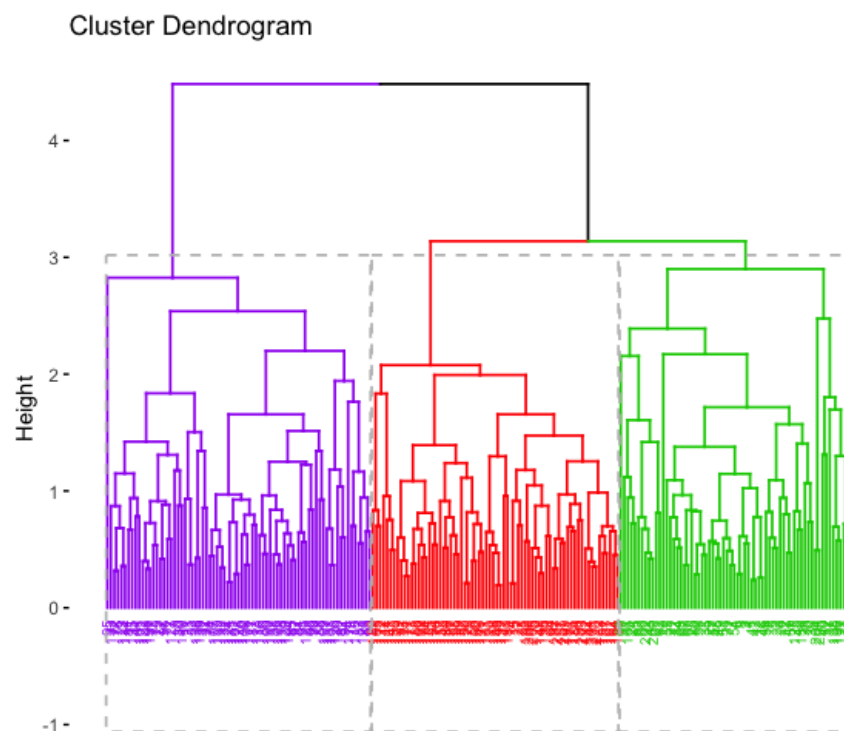
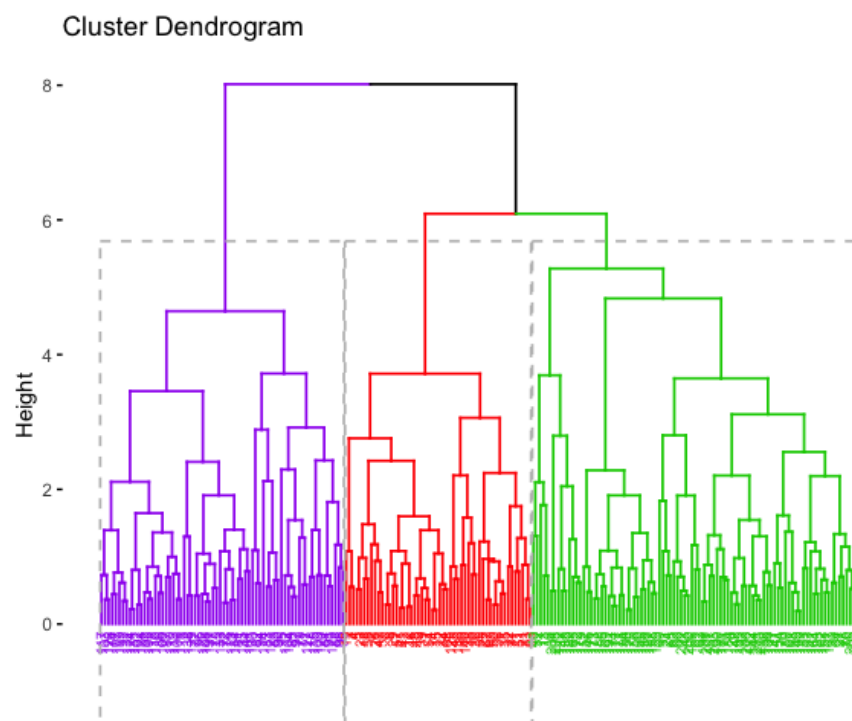


Figure 7: Cluster Dendrogram of Model 3.1 (Complete Linkage)



III. Interpretations and Recommendations

In this analysis, four models were created to cluster 210 seed samples into similar groups using measurements and calculations related to the seeds characteristics. The initial model created four clusters of sizes ranging from 30 to 65 and an explained variance of 74.7%. Using information from that model, the ideal number of clusters was identified as $k=3$, and a second model was created with 3 clusters. Model 2 was more uniform in size (groups ranging from 67 to 72), and the explained variance is 70.7%.

Looking at the visuals of the plots (Figure 3 and 4), Model 2 appears to be the better model because it creates three distinct groups that are mostly unique from each other. While the explained variance was smaller in Model 2, it was only 4 percentage points lower, which is marginal and not worth adding another cluster to the model.

Two additional models were created using the hierarchical method, one model using average linkage and the other model using complete linkage (Figures 6 and 7). The two models are relatively similar to each other. The preferred model is Model 3 because the groups are more even in size, and using average linkage creates a model based on all the data points, not only the points farthest apart (as with complete linkage).

While the k-means and hierarchical methods created strong models at clustering seeds, we know they are not perfect models due to not creating equal groups of 70. (We know the data represents 70 samples of 3 types of seeds from the documentation accompanying the .txt file). Further tuning of the model is required to correctly cluster the data. This can be achieved by finding a different way of standardizing the data, eliminating variables that may not be helping the clustering, or further tuning the way the models are created.

References

Seeds. UCI Machine Learning Repository. (n.d.). <http://archive.ics.uci.edu/dataset/236/seeds>


```
> #Model summary
```

```
> km_model
```

K-means clustering with 4 clusters of sizes 51, 65, 30, 64

Cluster means:

	Area	Perimeter	Compactness	length_of_kernel	width_of_kernel	asymmetry_coefficient	length_of_kernel_groove
1	1.4464555	1.4352979	0.6515415	1.4258480	1.3299889	-0.1468576	1.4464642
2	-0.3094639	-0.3593419	0.4349161	-0.4671207	-0.1370311	-0.7916699	-0.7890334
3	0.4293031	0.4803988	0.2326622	0.4532082	0.4334718	0.1858475	0.4580621
4	-1.0395808	-1.0039833	-1.0701098	-0.8742446	-1.1238527	0.8339509	-0.5660058

Clustering vector:

[illegible]

Within cluster sum of squares by cluster:

```
[1] 86.31332 122.55172 49.82717 110.72485
(between_SS / total_SS = 74.7 %)
```

Available components:

```
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"    "size"         "iter"
[9] "ifault"
# Check results of the fit
```

```
> # Centers of each cluster for each variables
```

```
> km_model$centers
```

	Area	Perimeter	Compactness	length_of_kernel	width_of_kernel	asymmetry_coefficient	length_of_kernel_groove
1	1.4464555	1.4352979	0.6515415	1.4258480	1.3299889	-0.1468576	1.4464642
2	-0.3094639	-0.3593419	0.4349161	-0.4671207	-0.1370311	-0.7916699	-0.7890334
3	0.4293031	0.4803988	0.2329622	0.4532082	0.4334718	0.1858475	0.4580621
4	-1.0395808	-1.0039833	-1.0701098	-0.8742446	-1.1238527	0.8339509	-0.5660058

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Appendix B: Model Summary of Model 2

```
> km_model_k2
```

K-means clustering with 3 clusters of sizes 71, 67, 72

Cluster means:

	Area	Perimeter	Compactness	length_of_kernel	width_of_kernel	asymmetry_coefficient	length_of_kernel_groove
1	-0.1407831	-0.1696372	0.4485346	-0.2571999	0.001643014	-0.66034079	-0.5844965
2	1.2536860	1.2589580	0.5591283	1.2349319	1.162075101	-0.04511157	1.1892273
3	-1.0277967	-1.0042491	-0.9626050	-0.8955415	-1.082995635	0.69314821	-0.6233191

Clustering vector:

[illegible]

Within cluster sum of squares by cluster:

```
[1] 144.4586 139.5542 144.5954
(between_SS / total_SS = 70.7 %)
```

Available components:

```
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"    "size"         "iter"
[9] "ifault"
```

```
> km_model_k2$centers
```

	Area	Perimeter	Compactness	length_of_kernel	width_of_kernel	asymmetry_coefficient	length_of_kernel_groove
1	-0.1407831	-0.1696372	0.4485346	-0.2571999	0.001643014	-0.66034079	-0.5844965
2	1.2536860	1.2589580	0.5591283	1.2349319	1.162075101	-0.04511157	1.2892273
3	-1.0277967	-1.0042491	-0.9626050	-0.8955451	-1.082995635	0.69314821	-0.6233191